

The background is a vertical gradient from yellow at the top to orange at the bottom. Overlaid on this are several blue lines that form a series of interconnected, rounded rectangular shapes, resembling a circuit board or a stylized path. These lines frame the text elements.

AI

BASICS

&

BIG IDEAS

GRACE LIU

TABLE OF CONTENTS

Letter from the Author	1
1 Mapping it out	3
2 Back to the Future: The Birth of AI	8
3 Think Like a Machine: How Machines Learn	14
4 AI Today: Netflix	28
5 AI Today: Apple Siri	38
6 Misconceptions & Moralities	48
7 So ... Is AI going to take over the world?	51
8 The Future is Definitely AI	54
References	57

Dear Reader,

Congratulations on picking up this book! Starting is literally one of the hardest things — I'm proud of you!

Before we get started, I just wanted to confess something: when I first started digging into AI in high school, it felt absolutely hopeless. I didn't have any teachers guiding me and all the information online just felt so overwhelming. I didn't understand how different topics connected with each other, what exactly AI is (it's not just robots), and I thought AI was just a product of extremely advanced coding that only people who knew computer science could get. So, don't worry if you don't know anything about technology, math, or computer science — I was in the same place you were not too long ago.

This book contains all of the foundational things that I wish I knew starting out. You'll learn a bit about what exactly artificial intelligence (AI) is, the history behind it, some of the big ideas, and a few examples of AI in your daily life — hopefully enough to make you want to learn more. Although AI sounds intimidating, I promise it's not as complex once you get the basics. By the end, I hope that not only will you understand why we have AI and why it's so important, but maybe this book will make you want to dive deeper into the field.

Anyways, here are some examples of AI that serve us in our daily lives (and I bet you didn't even realize some of these were AI):

- Siri, Alexa, Google Assistant
- Snapchat filters, TikTok "For You" page recommendations
- Spotify's and Netflix's recommendations
- Thermostats (Nest)
- Self-driving cars
- ChatGPT (obviously)

And this is only the tip of the iceberg.

Happy reading,
Grace Liu

MAPPING IT OUT

Artificial intelligence is everywhere around us. I know that sounds a little daunting, but I'm almost positive you've encountered it in your day-to-day. One example of that is the streaming platform everyone knows and loves: Netflix.

At the core of Netflix's success is its sophisticated AI recommendation system designed to personalize content and enhance your experience — that's how you get your recommended movies and TV shows. Recommendation systems are also used in a lot of other places, such as the “Categories to explore” and “Related to items you've viewed” sections of Amazon, Tinder profiles, and Spotify's “Discover Weekly.”

But, how exactly does the AI in Netflix work? What makes it “AI?”

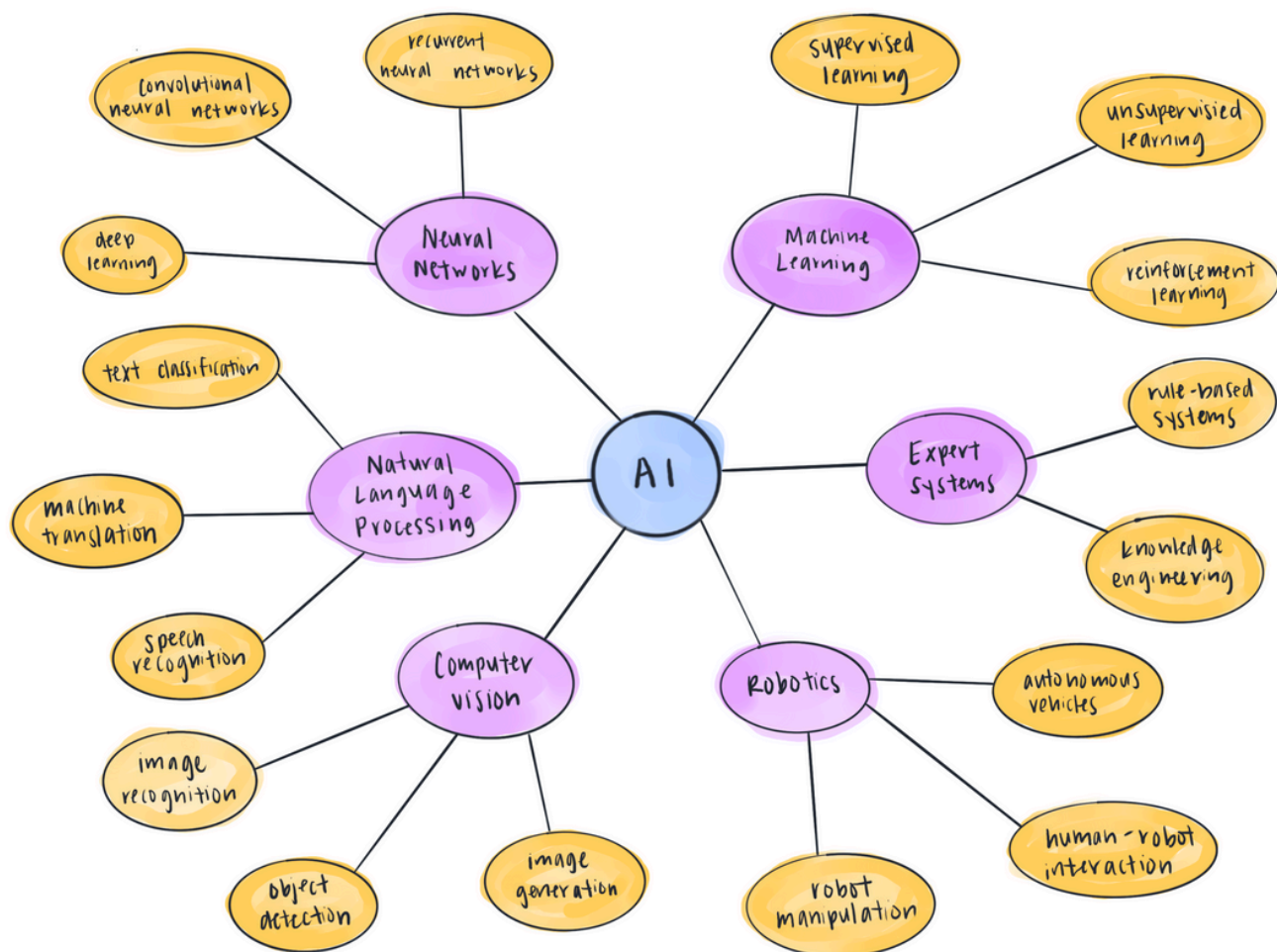
Before we dive into how artificial intelligence works, especially in daily tools such as Netflix, it's important to understand the different goals, or fields, of artificial

intelligence, and some of its approaches.

WHAT ARE AI SYSTEMS BUILT UPON?

Data structures and algorithms create the framework for data/information so that AI systems can effectively use the data that is given to them. A data structure organizes data in a “structured” format so that AI can use it in a more efficient manner. Think of it as a library, where books are arranged in specific sections, shelves, and genres, making it easy to find exactly what you are looking for — data structures do the same thing when organizing data. An algorithm is a set of step-by-step instructions, kind of like a recipe, that tells the AI system what to do. Together, these two principles ensure that data is organized and used correctly and efficiently, allowing AI systems to do their thing as best as possible to achieve the goals of AI.

Now, here are some of the main branches of artificial intelligence.



Machine learning: create AI models that learn from data over and over again so their performance improves as they learn more and repeatedly. It's like learning to play the piano: An AI is given a piece of sheet music (also known as "training data") and they play it. The piano teacher (feedback) tells the AI what it played right and wrong, and then the AI adjusts and tries again, over and over until the AI plays the piece perfectly.

Neural networks are one of the most common and furthest developed approaches to creating AI systems. These are technically a subset of machine learning but are inspired by the human brain (Hence, “neural.” Because neurons power the brain ... get it?). A network consists of a bunch of algorithms that recognize patterns in data. This approach is commonly used in computer vision and natural language processing, among others.

Deep learning is essentially multiple neural networks layered upon each other so they can identify more complex patterns in data (Just mentioning this one because it’s also one of the most studied and worked-on approaches).

Computer vision: aims to help computers see and understand images and videos the way that humans do. This includes identifying objects, people, locations — basically everything — and also creating images and videos. Computer vision can be used in robotics to help with visual navigation within its environment.

Robotics: robots. The self-driving cars that you have probably been seeing on Instagram and TikTok are robots, and also those cute, little food delivery robots.

Expert systems: systems that mimic how humans would make decisions. An expert system specializes in one thing, kind of how a bird expert specializes in birds while a car expert is all about cars. Some examples of these are medical diagnosis apps and customer service chatbots.

Natural language processing: techniques that will allow computers to understand human language. This includes translating languages, such as Google Translate, speech recognition when you talk to an Amazon Alexa, and when your email filters spam based on analyzing the text.

Again, this IS NOT a comprehensive map of everything in AI, just some of the biggest ideas that I recommend you should know starting out.

Next up: how was the concept of artificial intelligence created?

BACK TO THE FUTURE — THE BIRTH OF AI

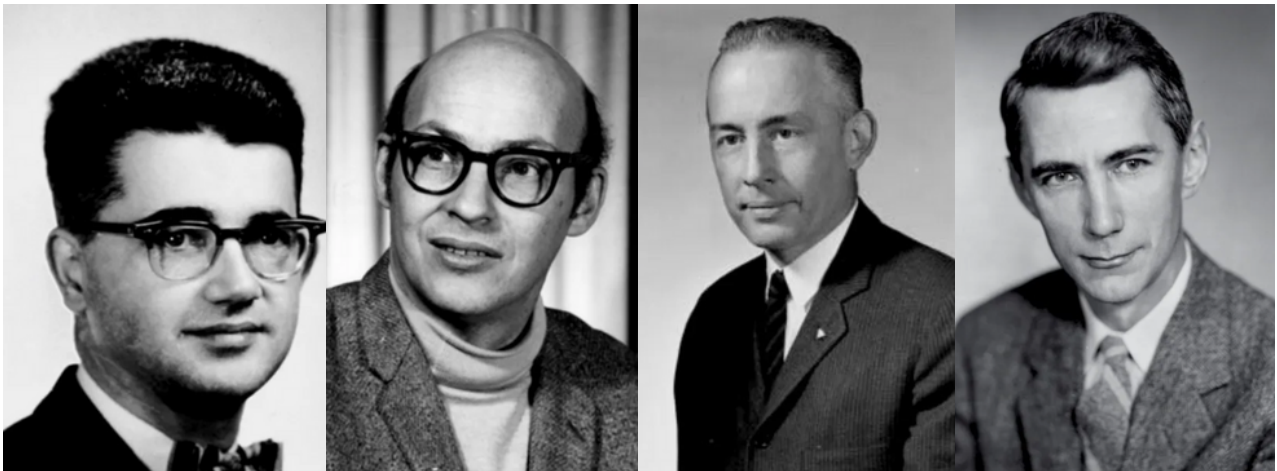
A long time ago in a galaxy far, far away, artificial intelligence was created.

No, I'm just joking. Did you really believe that? AI isn't alien (actually, I'm not too sure about that, you can ask space researchers that). On Earth, the concept of artificial intelligence was founded less than a century ago, in 1956, at Dartmouth College in Hanover, New Hampshire.

Everything started during the summer of 1956, when mathematician John McCarthy organized the Dartmouth Summer Research Project on Artificial Intelligence workshop, which brought together some of the brightest minds in cognitive science. McCarthy, along with cognitive and computer scientist Marvin Minsky, computer scientist Nathaniel Rochester, and mathematician and electrical engineer Claude Shannon, teamed together to discuss the possibility of “intelligent machines,” whatever that is.

“At the time, I believed if only we could get everyone who

was interested in the subject together to devote time to it and avoid distractions, we could make real progress,” — John McCarthy.



John McCarthy

Marvin Minsky

Nathaniel Rochester

Claude Shannon

(Photo credits: MIT Museum, Institute of Electrical and Electronics Engineers Inc)

The four of them ended up submitting a proposal to the Rockefeller Foundation to try and fund their “2-month, 10-man study of artificial intelligence.” The ambitious idea was that “every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it.” This was, and still is, absolutely crazy. Think about it: Back in the '50s, they had just invented the first hard computer disk, and they were already talking about artificial intelligence??

Anyways, their proposal was revolutionary (even though Rockefeller only ended up giving them half of the money

they requested) and many loved it, as evident in our present-day society. This was the start of artificial intelligence research.

SO, WHAT WAS THE GOAL OF AI?

The original goal of AI, according to McCarthy, was crazy ambitious: to build a fully intelligent machine in a decade.

Huh?? They wanted to create legit, naturally intelligent artificial intelligence by 1966? I know that the '60s were all about retrofuturism and stuff, but they had just invented satellites, the computer mouse, and L.E.D.s. You know what, good for them for dreaming big.

Anyways, their goal was founded on the idea that if they could use logic and computation to perfectly, identically, indistinguishably emulate any and all parts of human intelligence, they could create their self-thinking machine.

WHAT EXACTLY DID THEY TALK ABOUT AT DARTMOUTH?

Participants of the workshop didn't just decide that artificial intelligence was going to be a thing and call it a day. One of the biggest, and most important, things they did was name

this field of science.

To differentiate this new, innovative field from cybernetics, the term “artificial intelligence” was invented (cybernetics is the science of controlling things using technology). Funny enough, people ended up not liking this name later on because the goal was genuine, independent intelligence, not something artificial.

Minsky created the phrase “suitcase word” for concepts such as intelligence, thinking, and cognition, among many others. This is because each concept is like a suitcase, with a bunch of different meanings. For example, you can say that intelligence is binary: something is or is not intelligent. Or, you can say intelligence is on a continuum: one thing is more intelligent than another thing. Or, you can say intelligence is multidimensional: someone can be book-smart, but not street-smart. Regardless, experts in AI have ignored such meanings generally, instead focusing on distinguishing AI research into two distinctions: scientific vs. practical. [Scientific AI](#) research would be trying to understand exactly how intelligence works in human biology and replicating it in computers. [Practical AI](#) research would be creating computer programs capable of performing tasks as well as, or better than, humans, without considering if these programs are thinking in the same way humans do.

Numerous ideas for approaching AI research were discussed, and people were often disagreeing and arguing (unsurprisingly. I mean, they're scientists. They're all stubborn). Some people advocated for math and deductive reasoning to replicate how humans think, others supported statistics and probability to account for human uncertainty/unpredictability, and utilizing biology and psychology to create programs that replicated the brain was also debated. There were many possible avenues debated but, spoiler, biology/psychology/neuroscience became one of the most popular approaches for achieving AI today (in the form of neural networks).

Topics that were discussed included abstract concepts and reasoning, natural language processing, neural networks, machine learning, and even creativity (but don't worry about these yet). Essentially, the foundations, the big ideas, and the general goals were created. Obviously, no intelligent machine or robot was created from that one summer, but a young, powerful, hopeful optimism was born.

WHAT ARE THE PRESENT GOALS OF AI?

As you know, we have not reached the original Dartmouth goal in 2024. Today, artificial intelligence is defined as a field of computer science that focuses on doing things and solving

problems the way that we humans do. That means we're trying to replicate human learning, problem-solving, and recognizing faces, in machines. Although it's not too far off from the original goal, the overarching goal/purpose for artificial intelligence is to improve the efficiency of daily human life and to come up with and brainstorm solutions for problems like a human would.

Now that you know what artificial intelligence is, why people are so obsessed with it, why it's getting so big, some of the big ideas, and why it's so important, we can get into the nitty gritty of it!

Next up: how do machines learn?

THINK LIKE A MACHINE: HOW MACHINES LEARN

AI TODAY — NETFLIX

Going back to Netflix, we can now talk about what makes Netflix “AI.” The streaming service’s recommendation system is made up of algorithms that use user data to create personalized recommendations for you so you can find a movie or TV show with minimal effort, as fast as possible. In fact, almost 80% of what people watch on Netflix comes from these suggestions. Netflix’s recommendation system redefined the entertainment industry, and such systems are crucial because they can handle massive amounts of data while making your experience as enjoyable as possible.

This chapter will delve a bit into the technical and aesthetic aspects of Netflix’s recommendation system, including machine learning algorithms, how user data is collected, how users such as us interact with the system, and ethical considerations.

A BRIEF BACKGROUND ON NETFLIX

Netflix started out less than 30 years ago in 1997 as a DVD-by-mail service. By 1998, they launched Netflix.com, offering less than 1000 DVDs but with perks like no due dates, late fees, or monthly limits—customers just got DVDs in their mailboxes. As the 2000s kicked off, Netflix rolled out a personalized movie recommendation system using past ratings to suggest new favorites, laying the groundwork for today's sophisticated algorithms.

The recommendation system was known as Cinematch, and was a collaborative filtering algorithm. Collaborative filtering stands as one of the most successful methods to develop recommendation systems, which uses a group of people's preferences to predict what other users might like. In 2001, Netflix introduced a five-star rating system to help make their predictions more accurate.

By 2002, Netflix added three more algorithms to Cinematch to narrow down on movies to recommend. A dynamic store algorithm checked local DVD availability to speed up delivery. Metasims collected movie details like plot and cast, while a search algorithm revealed what users were looking for.

Over time, the recommendation system started becoming more personalized, especially after introducing Profiles in

2005, allowing users to create different profiles for different users and/or different moods.

Netflix wanted to grow fast and improve its recommendation system, however, the company was still a startup with not a lot of resources. In October 2006, Netflix presented the Netflix Prize, an open competition to create an accurate, efficient collaborative filtering algorithm better than Cinematch for its recommendation system. Cinematch was 9.6% more accurate than simply predicting movie averages.

The turning point was in 2007, when Netflix went completely online and introduced streaming.

The Netflix Prize ended in 2009 when the winning algorithm had a greater accuracy of 10.06%.

At the beginning of the 2010s, Netflix became available on mobile devices, and had a streaming experience dedicated to kids. Throughout the rest of the decade and into the present, Netflix has been continuing to improve its recommendation system into what we know and love today.

SO, HOW DOES IT WORK?

I know that the concept of the recommendation system doesn't seem that complicated, and you're right. At least the general idea of it. There are two main aspects: [data collection](#) and [algorithms](#) (just a reminder: algorithms are simply instructions).

DATA COLLECTION

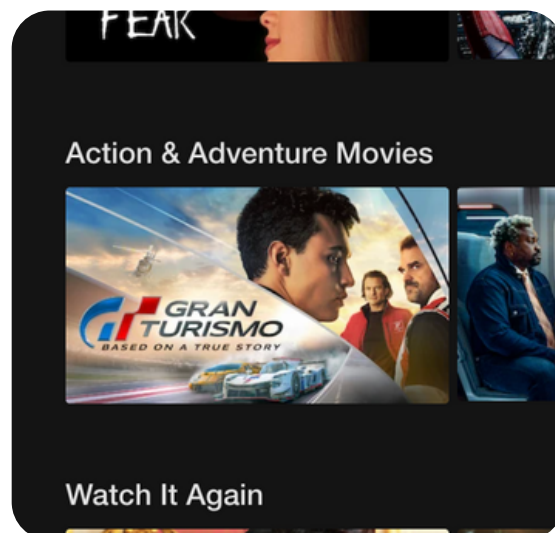
Netflix's present-day recommendation system collects many pieces of information from its users, some so niche that you wouldn't even have guessed mattered. This includes viewing history, the time spent on each title, search queries, and the viewing behavior of similar users, to name a few. Content data such as titles, genres, categories, actors, directors, release years, etc. are taken into account. External factors such as the time of day a user is watching, the devices that are being used to stream Netflix, membership duration, and the length of time spent watching are also considered in the decision-making process. However, demographic information is not considered due to privacy concerns, to avoid stereotyping and bias. Plus, behavior is more predictive anyway.

All of this data is then put through machine learning algorithms.

MACHINE LEARNING ALGORITHMS

Today, Netflix has still stuck with the machine-learning technique of collaborative filtering. This method looks at users' past behavior to suggest movies and shows. Think of it as Netflix having a large toolbox full of different tools — algorithms — to cover as much ground/information as possible to help provide you with recommendations.

One of these tools is the personalized video ranker algorithm which picks movies for you in different genres to create the “Survival Horror” and “Documentaries” rows. So, the horror movies you see might be different from what your friend sees.



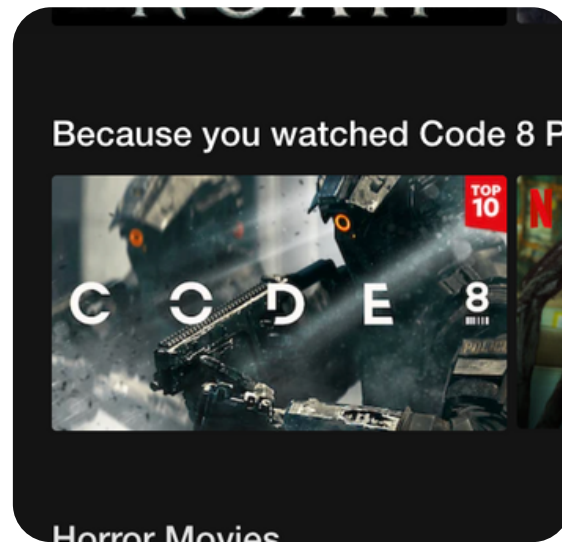
The Top N video ranker gives you the “Top Picks” row, providing you with the best suggestions for you from all of Netflix.



The Trending Now ranker algorithm balances what's currently popular with what you like, such as more rom-coms around Valentine's Day.



The video-video similarity algorithm creates the “Because You Watched” rows. While it's not super specifically personalized to you, it helps suggest similar shows or movies based on what you've watched.



Then, there's text mining which digs into movie titles, genres, actors, and more to make recommendations.

Last but not least, a page generation algorithm tool compiles all of these recommendations together on your screen in a way that's just for you, only for you.

DEEP LEARNING TECHNIQUES

Beyond machine learning, deep learning is also used for the recommendation system to create better recommendations. Netflix uses two main types of deep learning models: the bag-of-items approach and the sequential model.

For the bag-of-items approach, imagine you're dumping all the movies and TV shows you've watched into a bag without

caring about organizing them — it doesn't care when you watched them. This approach helps Netflix suggest shows by finding patterns in your watch history, using something called matrix factorization, a method that helps figure out what you might like based on what you and others have watched.

Sequential models, on the other hand, consider the order of what you watch. In Netflix, it predicts the next movies or TV show you will interact with based on the information of what you're currently viewing.

WAIT ... IT ALSO PERSONALIZES HOW NETFLIX LOOKS?

This may be a shock to you — I know I never noticed this — but the Netflix recommendation system also changes up the design of the platform (the user interface (UI)). So, when you're browsing, not only are the genre rows unique to you, but the artwork of the thumbnails of each movie and TV show is also personalized to you. For example, a user who mainly watches action movies will see a different thumbnail for a movie than a user who mainly watches musicals. The custom UI maintains user interest, low turnover rates, and increases the amount of time spent watching. A positive user experience then encourages users to use Netflix, and to stay.

ETHICS AND PRIVACY CONCERNS (SURPRISE, SURPRISE)

Since Netflix is taking all of this information from users like us, there are privacy concerns that come with it. Even though Netflix is in line with data regulations, there are risks of leaking data and questions about how data is stored and secured.

There are also ethical concerns with the recommendation system as it can perpetuate biases and stereotypes, potentially due to user profiling. This issue is part of the broader challenge within the entire field of AI.

While streaming on Netflix is way better for the planet than using DVDs, it still has its own environmental issues. Data centers and networks that help stream your favorite shows produce about 1% of the world's energy-related carbon emissions, which adds to global warming. The tech behind Netflix, like its recommendation system, uses a lot of energy too, creating a significant carbon footprint because of all the electricity needed both by Netflix and its users.

But, it's comforting to know that Netflix is working on reducing its environmental impact. As of 2022, they achieved

a net zero carbon footprint, meaning they've balanced out their emissions by doing things like improving energy efficiency in their data centers. These upgrades make streaming more energy-efficient than many other industries and help the planet. Plus, a smarter recommendation system means you browse less, saving even more energy.

While this AI system is massively successful, there is still the challenge for it to consider individual preferences. This means global and cultural backgrounds and the general unique and diverse nature of human behavior. Regardless, Netflix has made a revolutionary impact in our everyday lives and, come on, I know you're grateful for this little piece of AI in your daily life.

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READ ME.

ChatGPT. Self-driving cars. Robots taking over the world. This is what you may think of when you think of artificial intelligence (AI). But AI is also Snapchat recognizing your face for filters or your Netflix recommendations – daily stuff that we don't think twice about.

The world of AI is vast but – I know that seems daunting – it's not too complicated to understand the basics. In this beginner's guidebook, I, someone who tried to self-learn this field in high school, compiled some of the basics and big ideas that I wish I had in one place before trying to get into the complicated stuff. I won't bombard you with complex jargon, computer science algorithms, or random math. I'm here to let you know what exactly AI is, and the purpose and goals of this field. Simply.

“The most captivating intro AI book there is on the market!” – *The Daily Nexus*

“This book really understands the wants and needs of the younger generation when it comes to learning about AI,” – *Forbes*

“Enlightening ... A must-read book that makes you yearn for more!” – *The New York Times*

